

CS-468: Software Quality Assurance & Testing

White-Box Testing — SIMPLE REVISION NOTES

Week 10 & Week 11 — short, simple, easy to memorize

1. Why White-Box Testing?

- Looks INSIDE the code, not just the spec.
- Does 2 jobs: (1) measures coverage of tests you already have, (2) designs NEW tests to fill the gaps.

REMEMBER IT AS: *Test → Measure → Fill the gaps.*

2. Test Coverage — The Basic Idea

$$\text{Coverage} = (\text{Items exercised} \div \text{Total items}) \times 100\%$$

- “Item” = whatever you're counting (statements, decisions, etc.)

⚠ 100% Coverage ≠ 100% Tested

- Only measures ONE angle — same coverage, different tests can still catch different bugs.
- Only sees code that EXISTS. Can't catch missing features.
- Missing spec item → caught by specification-based testing.
- Missing from spec entirely → maybe caught by experience-based testing.

REMEMBER IT AS: *Coverage checks what's THERE — never what's MISSING.*

3. Statement Coverage

$$\text{Statement Coverage} = (\text{Statements run} \div \text{Total statements}) \times 100\%$$

- Just checks: did this LINE execute? Yes/No.

Example (quick):

IF A > B THEN C=0 ENDIF → 1 test with A>B (e.g. A=12, B=10) = 100% statement coverage.

A	B	C	Lines Hit	Coverage
2	3	8	1,2,3,4,6	83%
0	25	50	1,2,3,4,6	83%
47	1	49	1,2,3,4,6	83%
20	25	70	1,2,3,4,5,6	100%

REMEMBER IT AS: *You need ONE test that makes the IF TRUE to touch every line.*

4. Decision Coverage

- Decision = IF / loop / CASE — anywhere with 2+ possible outcomes.

$$\text{Decision Coverage} = (\text{Outcomes exercised} \div \text{Total outcomes}) \times 100\%$$

Example (quick):

- Test 1: A=20, B=15 → C=-10 → TRUE branch only.
- Add Test 2: A=10, B=2 → C=6 → FALSE branch.
- Together = both branches covered = 100% decision coverage.

☆ **GOLDEN RULE:** 100% Decision Coverage → always gives 100% Statement Coverage. 100% Statement Coverage → does NOT always give 100% Decision Coverage.

REMEMBER IT AS: Decision Coverage = the harder exam. Statement Coverage = the easy quiz. Passing the hard exam auto-passes the easy one — not the other way round.

5. Loops — While / Do-While / For

Loop	Check Condition	Body Guaranteed?
WHILE / FOR	Before body runs	No (may run 0 times)
DO-WHILE	After body runs	Yes (always ≥ 1 time)

REMEMBER IT AS: WHILE/FOR = look before you loop. DO-WHILE = leap first, look after.

6. Case Study — Gold Card / Airline Upgrade

Passenger with a gold card → may skip straight to upgrade. No gold card + full economy + late check-in → may get bumped. 10 numbered boxes, 4 decision points (boxes 1, 2, 6, 8).

(a) 5 Tests = Maximum Decision Coverage

Card	Business Class	Economy	Path
Gold	Not Full	Don't Care	1,2,4,5
Gold	Full	Not Full	1,2,3,5
Not Gold	Don't Care	Not Full	1,6,7,5
Not Gold	Not Full	Full	1,6,8,9,5
Not Gold	Full	Full	1,6,8,10

(b) 3 Tests → Statement Coverage Check

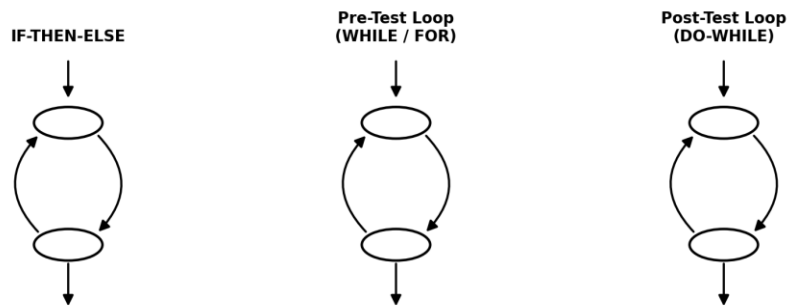
- Test 1: Gold, upgraded → path 1,2,4,5 → $4/10 = 40\%$
- Test 2: Not gold, economy → path 1,6,7,5 → $4/10 = 40\%$
- Test 3: Bumped → path 1,6,8,10 → $4/10 = 40\%$

REMEMBER IT AS: 5 tests = full branch coverage. Any smaller set (like 3) gives partial coverage — always check paths against total statements (10).

7. Control Flow Graph (CFG)

- A flowchart's skeleton — shows ONLY decisions, skips plain lines.
- **Node** = decision point OR where a structure re-joins the main flow (e.g. ENDIF).

- **Edge** = line joining two nodes.
- **Region** = closed shape formed by nodes + edges.



If-Then-Else | Pre-Test Loop (While/For) | Post-Test Loop (Do-While)

- If-Then-Else: splits into 2 branches, then rejoins.
- Pre-test loop: check FIRST, may skip body entirely.
- Post-test loop: run body FIRST, check after.

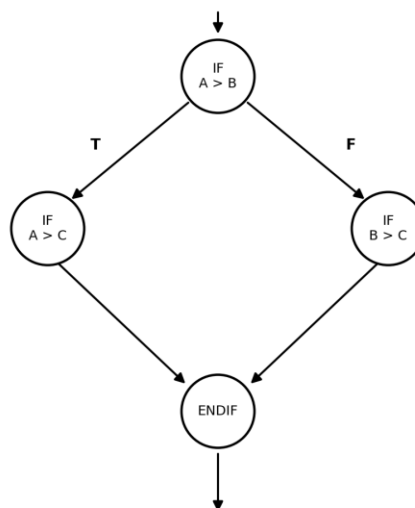
REMEMBER IT AS: *Any code = built from just these 3 shapes.*

3 Steps to Draw a CFG:

1. Find every decision (ignore plain lines).
2. Add 1 node per decision.
3. Expand each node with its matching subgraph shape.

Quick Example — MaxAndMean

Nested IFs: IF A>B → (IF A>C) else (IF B>C) → both merge into one ENDIF.



Both nested IFs converge on the same ENDIF.

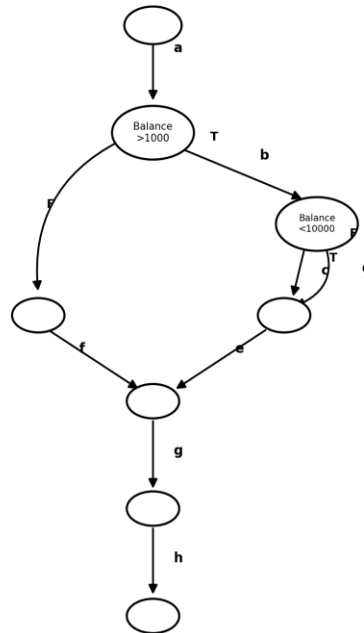
8. Hybrid Flow Graph

- = CFG + a letter on every edge.
- A path through the code becomes a simple code, e.g. 'abeh'.

REMEMBER IT AS: *Think of it as a treasure map with lettered roads — a path is just the sequence of letters you walk through.*

Quick Example — BestInterest

Balance > 1000 branches into 2 real paths: Balance 1,000–10,000 (path 'abcegh') and Balance > 10,000 (path 'abdfgh').



Paths 'abcegh' and 'abdfgh' — 2 tests give 100% statement coverage.

Answer: 2 tests (e.g. Balance=£5,000 and Balance=£20,000) = 100% statement coverage.

9. Same Case Study → CFG + Hybrid Version

- 4 decision diamonds (gold card?, business class full? ×2, economy full?) → 4 CFG nodes.
- Each node = if-then-else shape → True/False edges rejoin at their own ENDIF.
- Label every edge with a letter for the hybrid version → name any test's path, just like 'abcegh'.

10. Formula Cheat Sheet

Coverage = (items exercised ÷ total items) × 100%
 Statement Coverage = (statements run ÷ total statements) × 100%
 Decision Coverage = (outcomes run ÷ total outcomes) × 100%

11. Exam Questions — Quick Answers

4. **Tests for 100% decision coverage?** → Cover every True AND False at every decision point.

5. **Tests for 100% statement coverage?** → Fewer needed — just touch every line at least once.
6. **Coverage from ONE input?** → Trace it, list lines/branches hit, apply formula.
7. **Generate data for max statement coverage?** → Smallest set of inputs touching every line.
8. **Generate data for max decision coverage?** → Every branch driven True once AND False once.

12. ⚡ One-Page Cheat Sheet (Last-Minute Recall)

Term	Remember it as
Test Coverage	$\text{items exercised} \div \text{total items} \times 100$
Statement Coverage	Did the LINE run?
Decision Coverage	Did BOTH branches (T/F) run?
100% Coverage	$\neq 100\%$ tested — only shows what's THERE
Decision vs Statement	100% Decision → 100% Statement (not reverse)
WHILE / FOR	check first — may run 0 times
DO-WHILE	run first, check after — always ≥ 1
CFG Node	decision point / re-join point
CFG Edge	line joining 2 nodes
CFG Region	closed area of nodes+edges
Hybrid Flow Graph	CFG + lettered edges = named paths
Gold Card Case Study	10 statements, 4 decisions, 5 tests = max coverage